Reg. No. | | | | | | | | | | | [14CE64]

SIXTH SEMESTER B.TECH EXAMINATION, JUNE – 2018

ESTIMATING COSTING & SPECIFICATIONS

Time: 3 Hours **Maximum Marks: 70**

Instructions:

1. Part-A is compulsory which carries 28 marks. Answer any 3 full questions from Part-B which carries 14 marks each.
2. Missing data may be suitably assumed.
3. Follow the written dimension do not scale the drawing.

PART - A

1. The Fig 1 shown detail of an office building. Work out the quantities for the following items using centre line method. **28**

i. Earth work in excavation in foundation.

ii. Lime concrete in foundation.

iii. I-class brick work in foundation and plinth in lime mortar.

iv. I-class brick work in superstructure in lime mortar.

PART - B

2. a. What is an Estimate? Explain the necessity of Estimate. **07**

b. Explain different types of Estimate of buildings. **07**

3. Estimate the cost of RCC roof slab in cc 1:2:4 over a room of internal size 3×6m, also calculate the quantity of materials required. Given slab thickness 12cm. **14**

Steel reinforcement:

Main Steel : 12mm dia @ 12cm c/c

Distribution steel : 6mm dia @ 18cm c/c

Cost of R.C.C. 1:2:4 Rs. 675/- cu.m

Cost of steel bars Rs. 515/- per quintal

4. The plan and section of the "septic tank" with soakpit is given in Fig 4. Workout the quantities and cost of the following item of work: **14**

i. Earthwork in excavation @ Rs. 250 /m^3.

ii. First class brick work in CM 1:3@ Rs. 7600/m^3.

5. Estimate the cost of Earthwork for a portion of length 200metre of a road in an uniform ground, the heights of banks at the two ends being 1.00m and 1.60m. The formation width is 10m and side slopes 2:1 (H:V). Assume that there is no transverse slope. Calculate by **14**

i. Mid-sectional Area method

ii. Mean-sectional area method

iii. Prismoidal formula

6. Write a detailed technical specification for any two of the following items. **14**

i. SSM in super structure in CM 1:4

ii. RCC (1:2:4) roof slab.

iii. Bed concrete in foundation in cc (1:4:8)

7. Write a short notes on the following: **(3.5×4=14)** **14**

i. Administrative sanction and technical sanction

ii. Earnest money deposit

iii. Valuation money deposit.

iv. Valuation and methods of valuation.

v. Tender and lease hold properties.

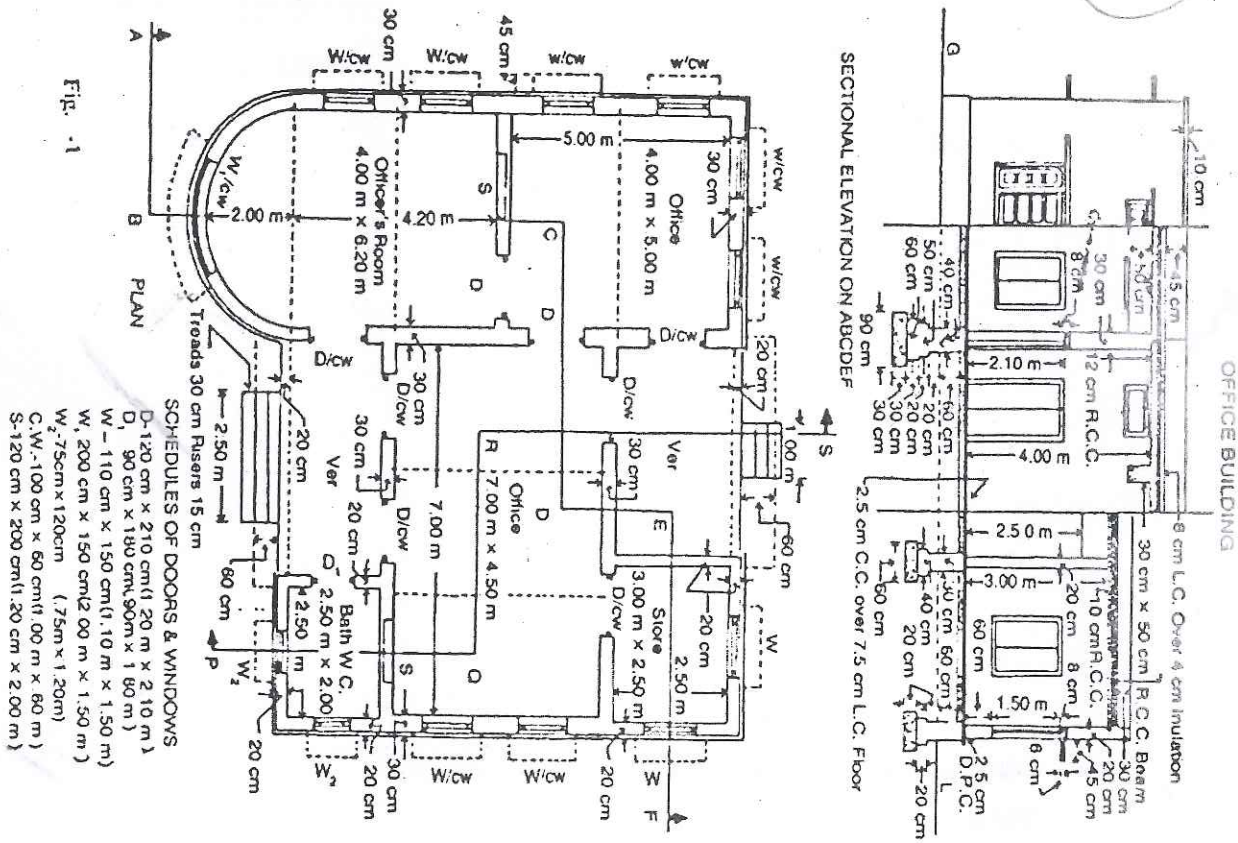
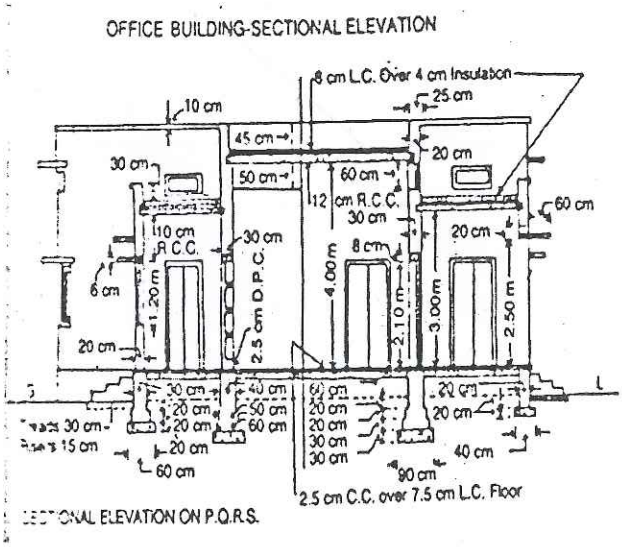


Fig. -1

Figure. 1.



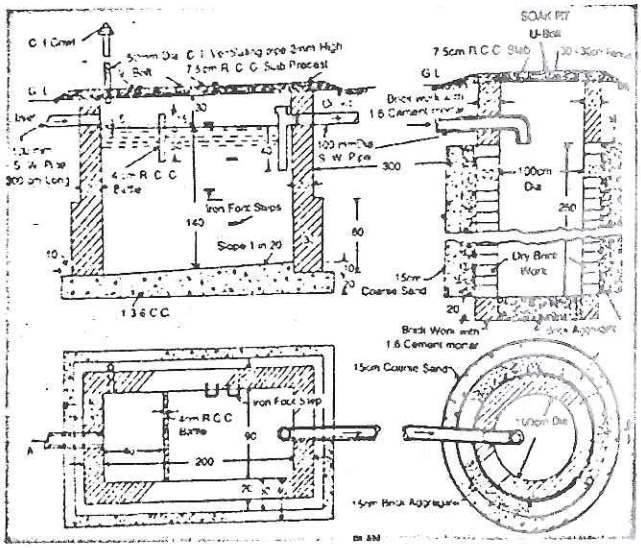


Figure. 4.